

7inch DSI Display

User Manual

1. Overview

- 800x480 resolution capacitive touch display screen
- Raspberry Pi MIPI DSI interface for direct output, plug and play, without the need for driver installation
- Support Raspberry Pi OS, the official system of Raspberry Pi, with 2-finger zoom for web pages
- Support Ubuntu-Mate/Kali, single touch;
- Support RetroPie
- Equipped with backlight brightness adjustment button

2. Parameter

Name	Explain
LCD Size	7inch
LCD Resolution	800x480
Screen Material	IPS
LCD Interface	RGB888
Module Interface	Raspberry Pi MIPI DSI
Refresh rate	60Hz
Touch screen type	Capacitive
Viewing Angle	Full perspective
Working Voltage	5V
Maximum operating current	510mA
Maximum Brightness	400cd/m²
Working Temperature	-20℃ -70℃
Module Size	164.89*102.00*12.25(mm)
Active Area	154.68*87.02(mm)
Packaging Size	212*149*54(mm)
Module Weight (net weight)	221g
Packaging Weight	1 330g

Troubleshooting:

I. No Backlight:

Check Power Supply:

1. If you've installed the Raspberry Pi on the back of the screen, check that the screen's spring pins are correctly aligned with the Raspberry Pi's pins and making full contact;
2. If using a jumper connection, check that it's correctly connected to the Raspberry Pi's 5V GND pin;
3. It's recommended to use the official power adapter. At least ensure the power adapter specifications meet official standards. For example, the Pi 5 uses a 5.1V 5A power adapter.

II. Backlight On but No Display:

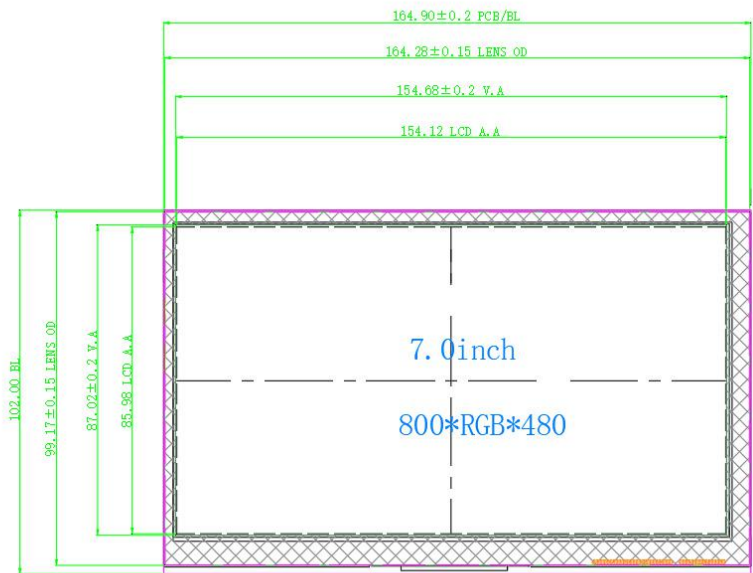
1. Check that the ribbon cable connection is oriented correctly and is fully inserted and securely locked in interface;
2. Use a multimeter to check if the Raspberry Pi's 5V GND pin is receiving power;
3. If you are using a Pi 5, please try another "DISP" port for testing;
4. Ensure you are using the official Raspberry Pi OS image with a desktop environment; other systems are not supported by plug and play.

If the above four methods still fail to get the screen to display correctly, and if you have a Raspberry Pi CSI camera (with a built-in ribbon cable), you can try connecting it to your Raspberry Pi and testing for image output. If there is image output, the ribbon cable is likely damaged; if there is no image output, the Raspberry Pi's interface is probably faulty. Alternatively, if you have a spare Raspberry Pi of a different model, you can try replacing it for testing.

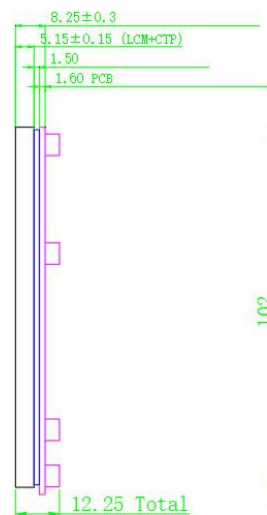
Finally, if after troubleshooting the above steps you still cannot get the screen to work properly, please contact us immediately. We will definitely provide you with a satisfactory solution.

3. Dimensional Drawing

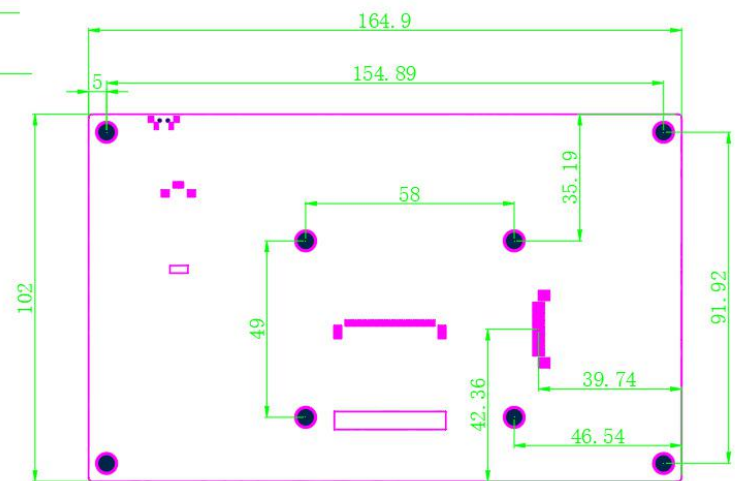
Front view



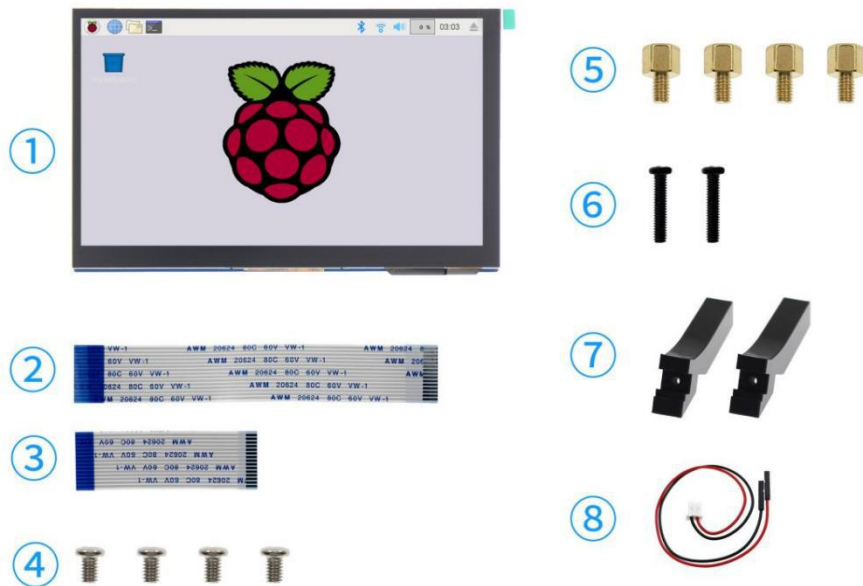
Side view



Rear view



4. Packaging List



Packaging List:

- | | |
|---------------------------------|-------------------------------|
| ① 7 inch LCD x1 | ⑤ Copper Pillar (M2.5x4+4) x4 |
| ② FFC-1.0-15P-10cm (reverse) x1 | ⑥ Screw (M2.5x12) x2 |
| ③ FFC-1.0-15P-5cm (reverse) x1 | ⑦ Support x2 |
| ④ Screw (M2.5x4) x4 | ⑧ 2P Wire Rod x2 |

5. Hardware Description

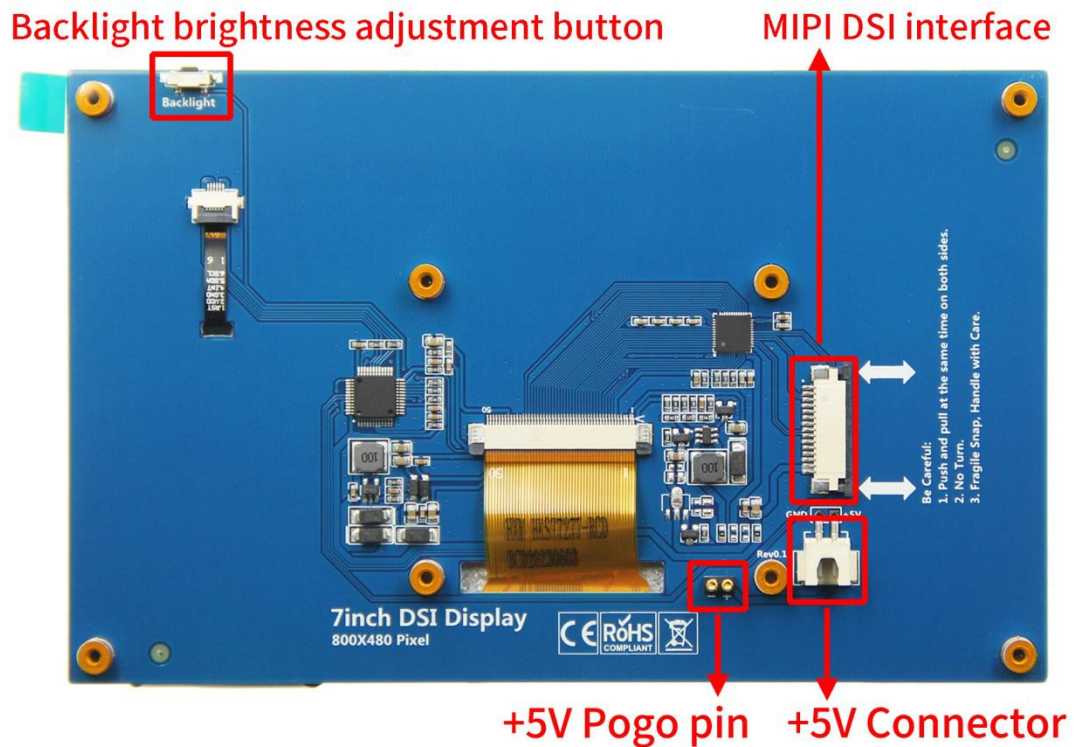


Figure 1. Hardware Interface

Detailed explanation:

- (1) **Backlight brightness adjustment button:** Short press backlight change 10%, long press 3 seconds to close backlight; short press to open backlight.
- (2) **MIPI DSI interface:** The specification is 15PIN with a spacing of 1.0MM; MIPI DSI interface for connecting Raspberry Pi through FPC flexible ribbon cable

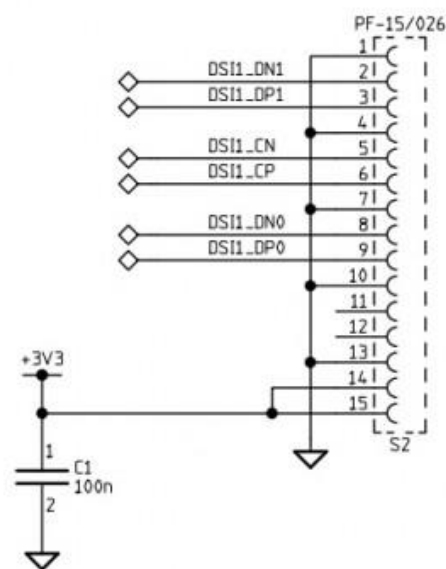
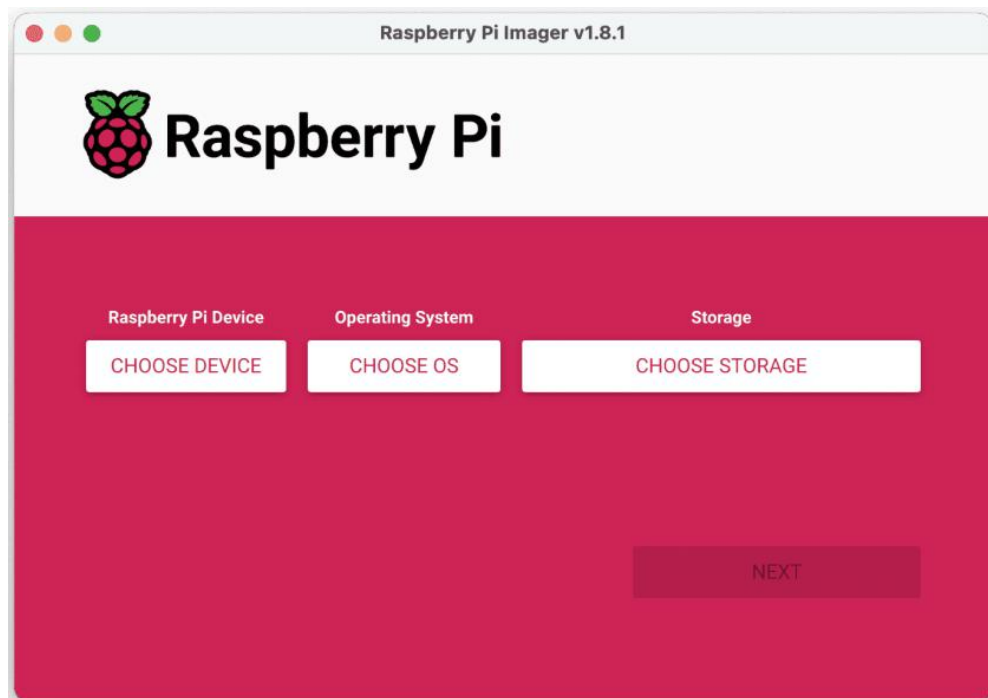


Figure 2. MIPI DSI 15PIN Pin Definition

- (3) **+5V Pogo pin:** Used to power from below the Raspberry Pi pin (Raspberry Pi needs to be fixed to the module with screws and copper pillars)
- (4) **+5V Connector:** Used to power from above the raspberry pie pin through the DuPont cable

6. How to use it

- (1) Install Raspberry Pi OS according to the official tutorial steps, and put the Micro SD card into the Raspberry Pi.



- (2) Fix the Raspberry Pi to the copper pillar with 4 screws (M2.5), then connect the FFC flexible ribbon cable to the screen (as shown in Figures 3 and 4), connect it to the Raspberry Pi MIPI DSI interface (as shown in Figure 5), and finally connect the Raspberry Pi to the power cord;

After the Raspberry Pi motherboard is powered on, the display and touch functions are normal.

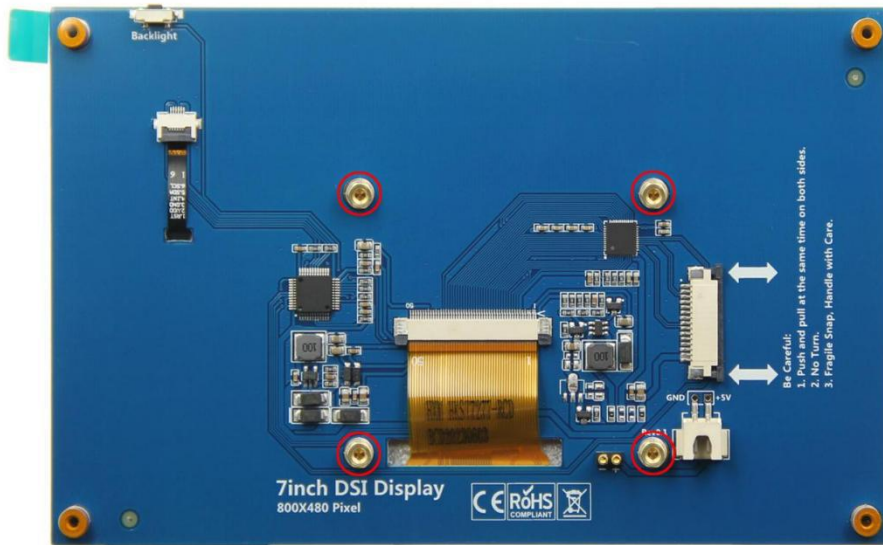


Figure 3. Installing copper pillars

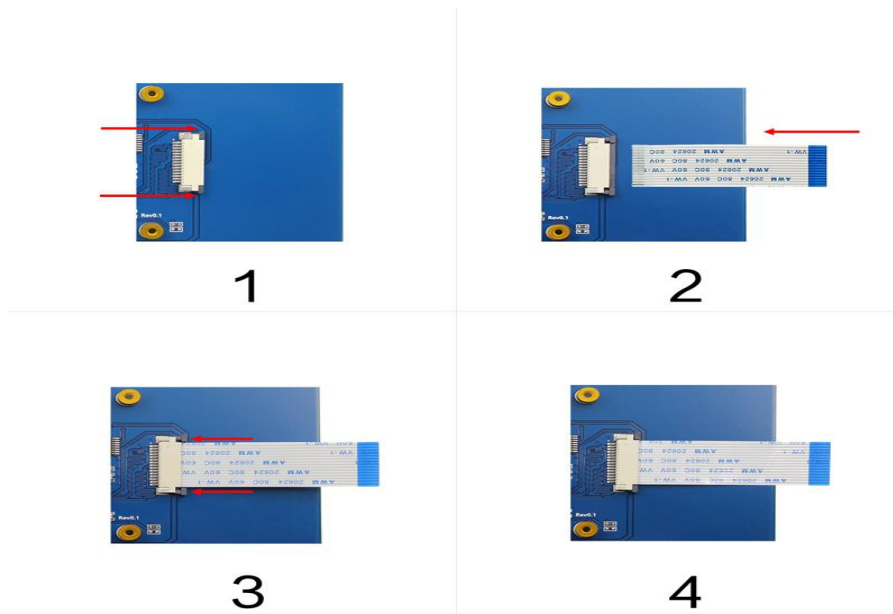


Figure 4. Connecting FFC

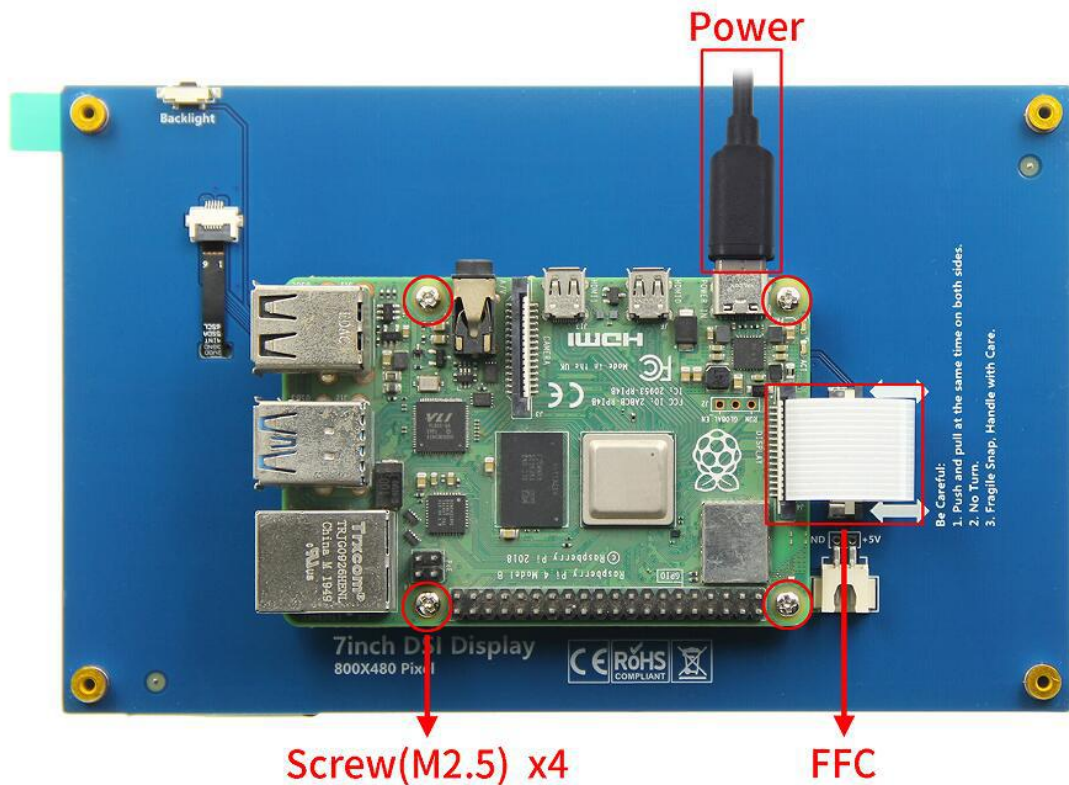


Figure 5 Connecting Raspberry Pi

Attention: The copper pillars, screws, and 15 PIN FPC cables mentioned above are all included in the product accessories.

How to connect Raspberry Pi, Method 2

Attention: When using this method of connection, a 2P custom DuPont cable must be used to connect pins 4 and 6 of the Raspberry Pi to power the module, otherwise it may damage the module!!!

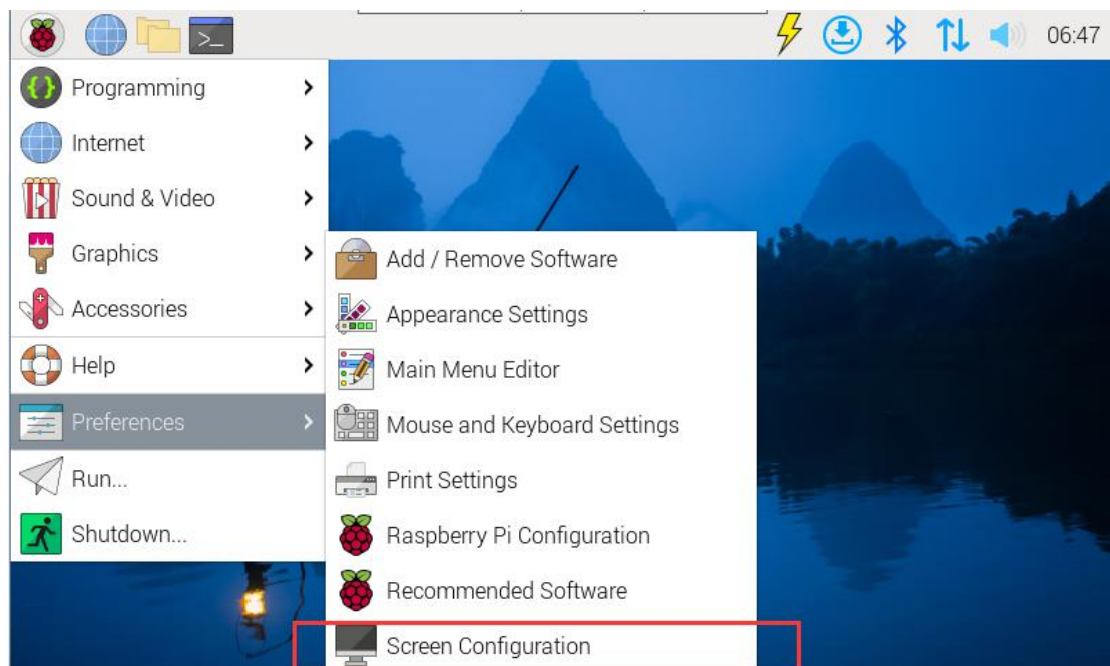
If you do not want the Raspberry Pi to be fixed on the module, you can also connect it using the following method:

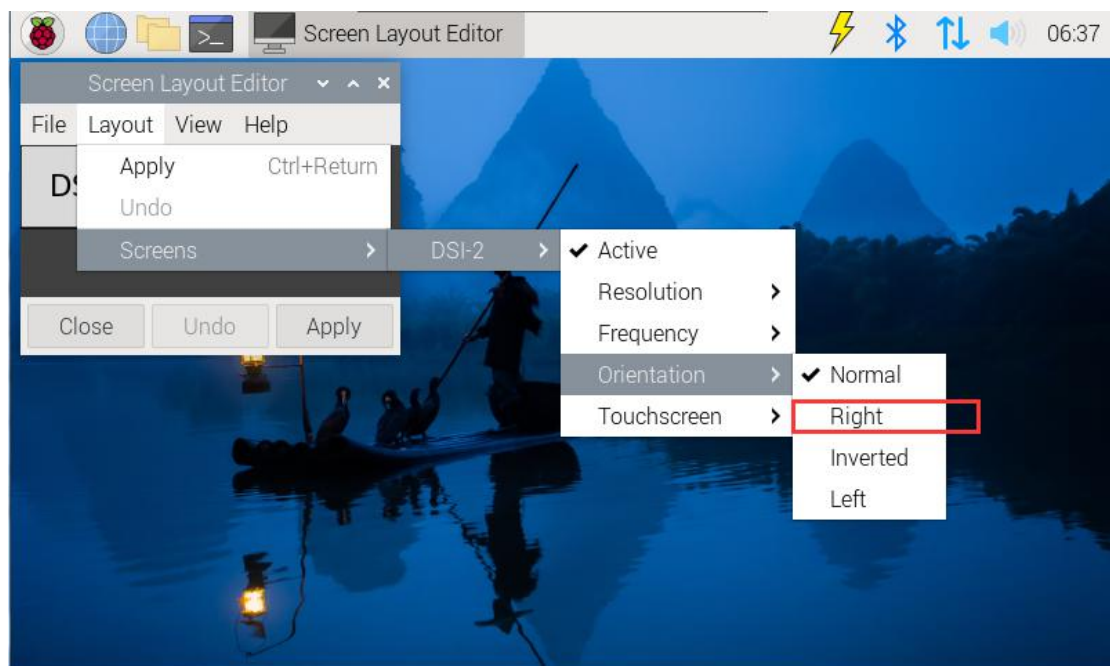
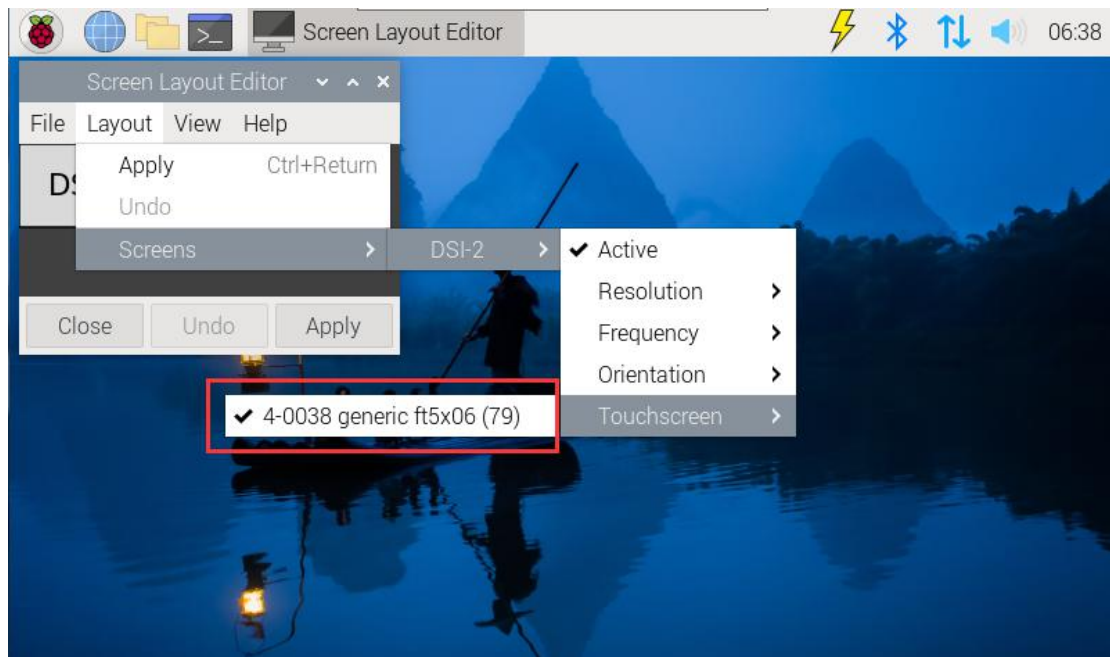
7. How to adjust brightness

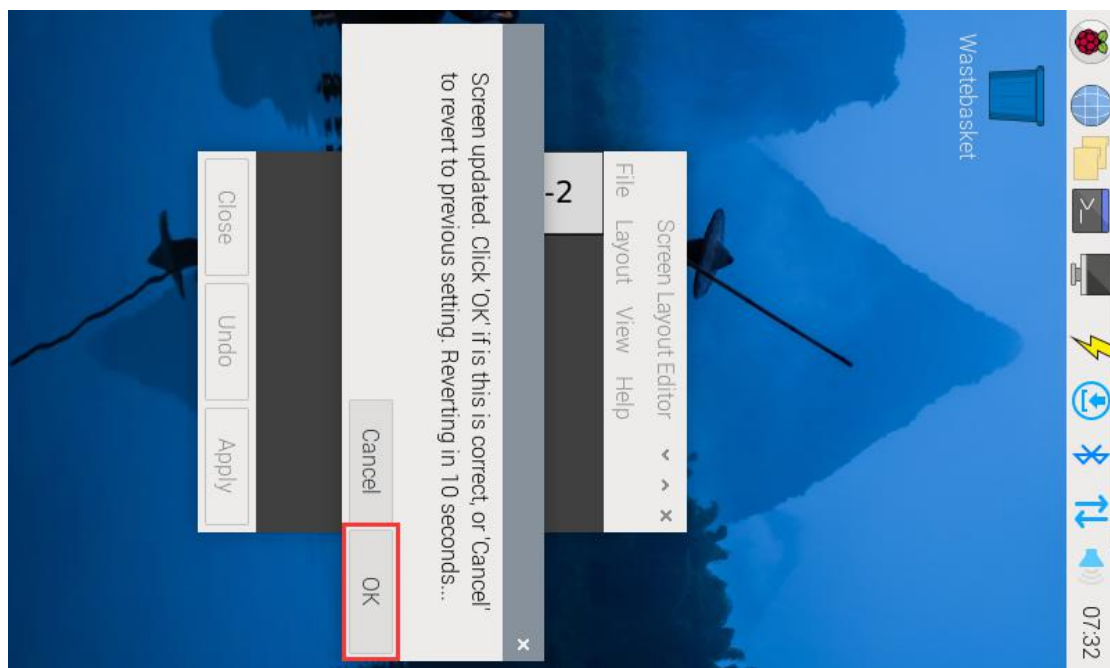
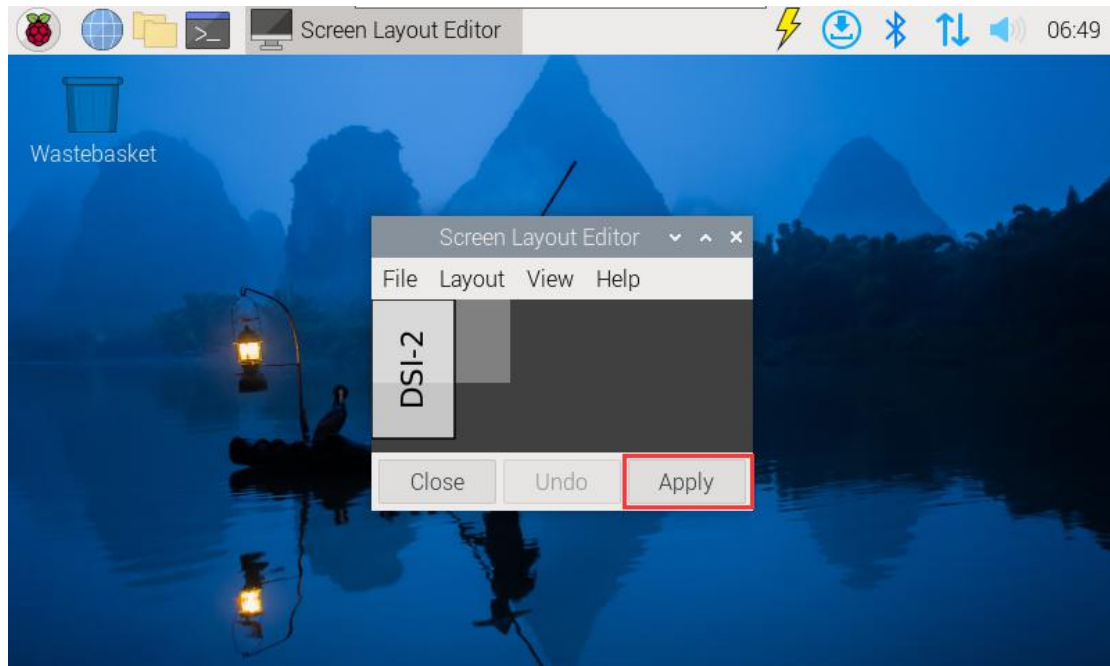
With each press of the button, the backlight brightness increases by 10%, reaching a maximum of 100%, and then cycles from 10% onwards; Long press for 3 seconds to turn off the backlight, then short press once to activate the backlight.

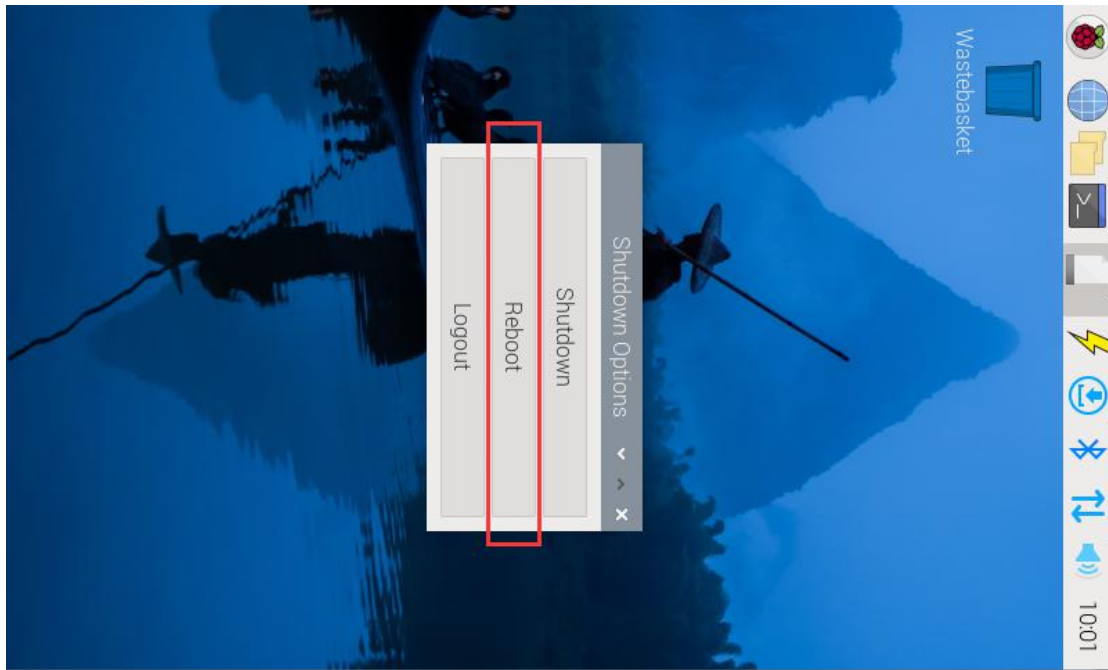
8. Display direction rotation

Following the method below, both the display and touch are rotated 90° clockwise.
(Only for official systems after 2023-12-05, older versions of the system are not applicable)









9. How to install a virtual keyboard

Execute the following command:

```
sudo apt-get install matchbox-keyboard
```